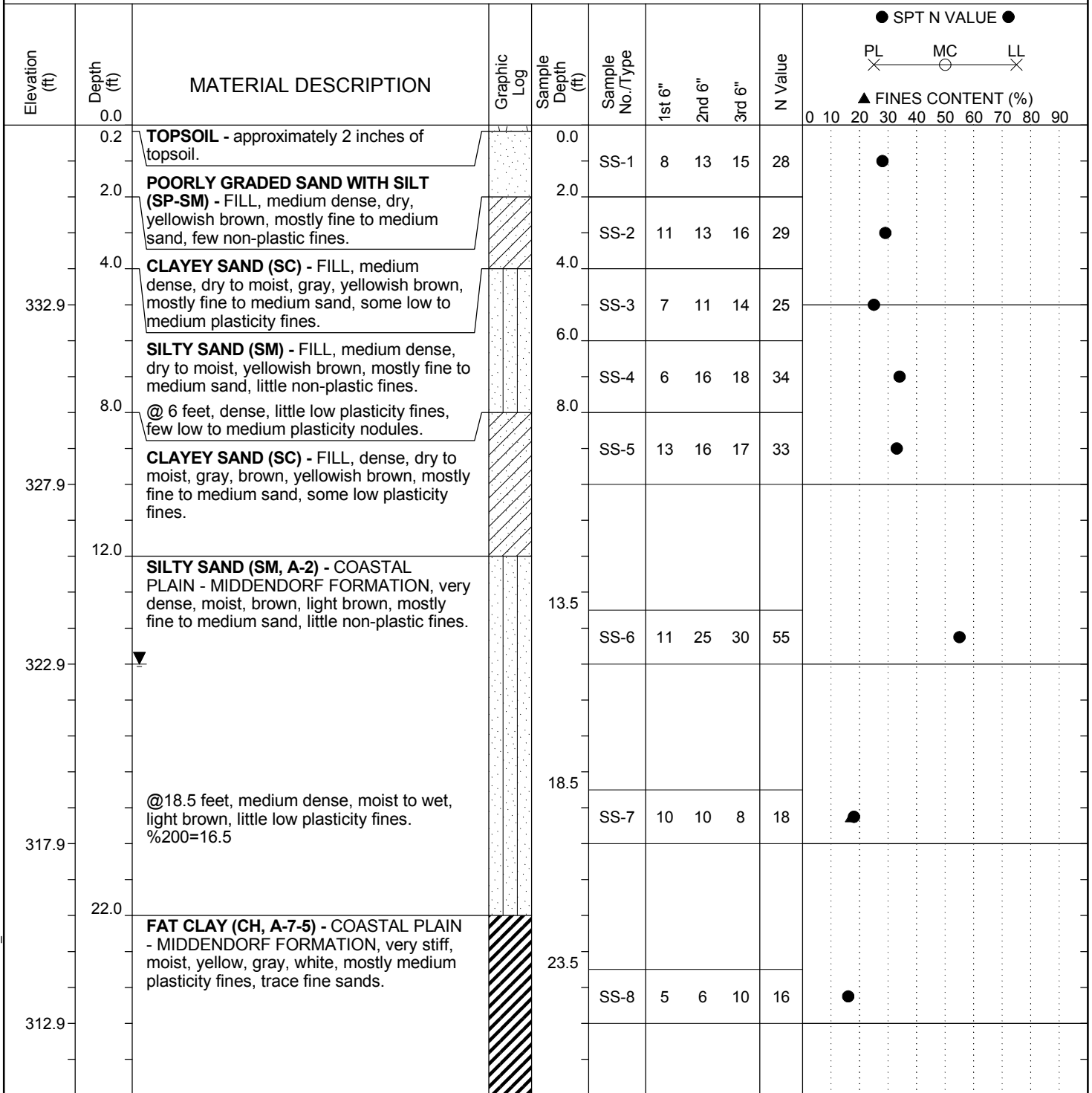


SCDOT Soil Test Boring Log

File No.:		Project No. (PIN):	1461-14-046	County:	Richland	Eng./Geo.:	S. Jones
Site Description:	I-77 Widening Design/Build Preparation					Route:	I-77
Boring No.:	B-01	Boring Location:	1215+74.4	Offset:	9.2' L	Alignment:	Mainline
Elev.:	337.9 ft	Latitude:	34.097467	Longitude:	80.956851	Date Started:	9/5/2014
Total Depth:	120 ft	Soil Depth:	120 ft	Core Depth:	N/A ft	Date Completed:	9/8/2014
Bore Hole Diameter (in):	4	Sampler Configuration		Liner Required:	Y (N)	Liner Used:	Y (N)
Drill Machine:	CME-45	Drill Method:	Rotary Wash	Hammer Type:	Automatic	Energy Ratio:	87%
Core Size:	N/A	Driller:	T. Miller	Groundwater:	TOB	RW, N/A	24HR 15 ft



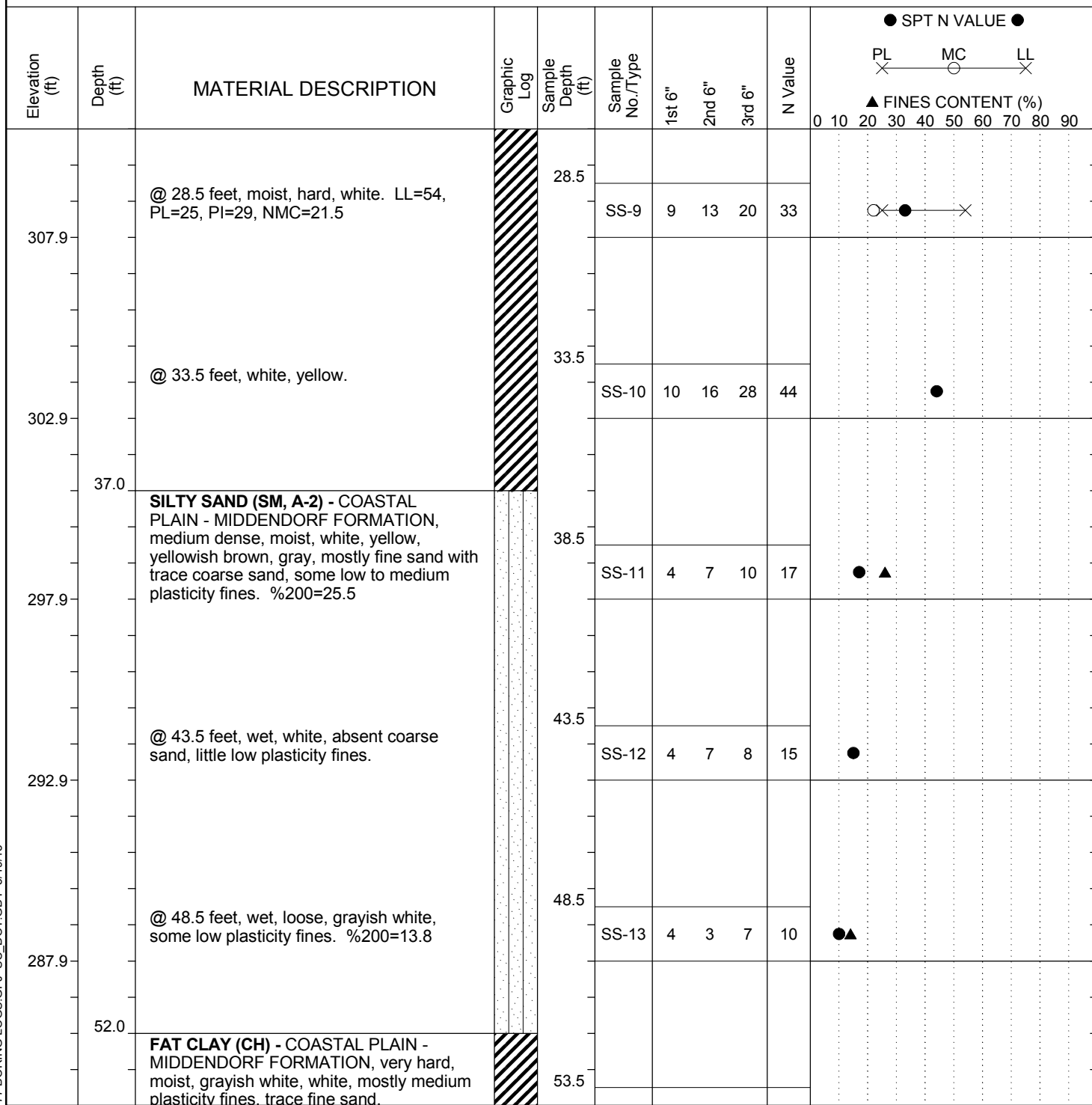
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SAMPLER TYPE		DRILLING METHOD	
SS - Split Spoon	NQ - Rock Core, 1-7/8"	HSA - Hollow Stem Auger	RW - Rotary Wash
ST - Shelby Tube	CU - Cuttings	CFA - Continuous Flight Augers	RC - Rock Core
AWG - Rock Core, 1-1/8"	CT - Continuous Tube	DC - Driving Casing	

SCDOT Soil Test Boring Log

File No.:		Project No. (PIN):	1461-14-046	County:	Richland	Eng./Geo.:	S. Jones
Site Description:	I-77 Widening Design/Build Preparation					Route:	I-77
Boring No.:	B-01	Boring Location:	1215+74.4	Offset:	9.2' L	Alignment:	Mainline
Elev.:	337.9 ft	Latitude:	34.097467	Longitude:	80.956851	Date Started:	9/5/2014
Total Depth:	120 ft	Soil Depth:	120 ft	Core Depth:	N/A ft	Date Completed:	9/8/2014
Bore Hole Diameter (in):	4	Sampler Configuration		Liner Required:	Y (N)	Liner Used:	Y (N)
Drill Machine:	CME-45	Drill Method:	Rotary Wash	Hammer Type:	Automatic	Energy Ratio:	87%
Core Size:	N/A	Driller:	T. Miller	Groundwater:	TOB	24HR	RW, N/A



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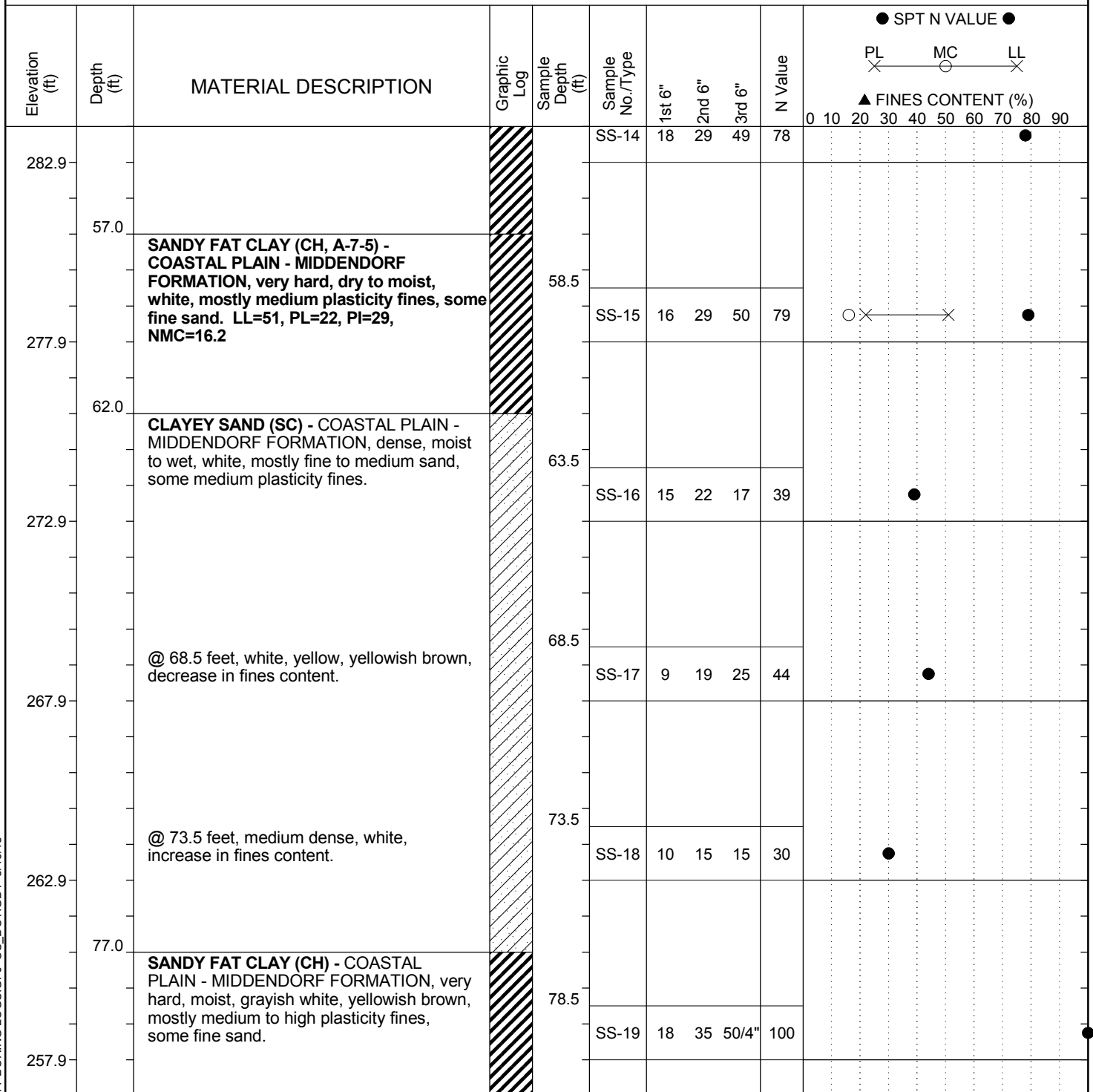
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SC_DOT_14-046 I-77 BORING LOGS.GPJ SC_DOT.GDT 5/19/15

SAMPLER TYPE		DRILLING METHOD	
SS - Split Spoon	NQ - Rock Core, 1-7/8"	HSA - Hollow Stem Auger	RW - Rotary Wash
ST - Shelby Tube	CU - Cuttings	CFA - Continuous Flight Augers	RC - Rock Core
AWG - Rock Core, 1-1/8"	CT - Continuous Tube	DC - Driving Casing	

SCDOT Soil Test Boring Log

File No.:		Project No. (PIN):	1461-14-046	County:	Richland	Eng./Geo.:	S. Jones
Site Description:	I-77 Widening Design/Build Preparation					Route:	I-77
Boring No.:	B-01	Boring Location:	1215+74.4	Offset:	9.2' L	Alignment:	Mainline
Elev.:	337.9 ft	Latitude:	34.097467	Longitude:	80.956851	Date Started:	9/5/2014
Total Depth:	120 ft	Soil Depth:	120 ft	Core Depth:	N/A ft	Date Completed:	9/8/2014
Bore Hole Diameter (in):	4	Sampler Configuration		Liner Required:	Y (N)	Liner Used:	Y (N)
Drill Machine:	CME-45	Drill Method:	Rotary Wash	Hammer Type:	Automatic	Energy Ratio:	87%
Core Size:	N/A	Driller:	T. Miller	Groundwater:	TOB	RW, N/A	24HR 15 ft



LEGEND

Continued Next Page

SAMPLER TYPE		DRILLING METHOD	
SS - Split Spoon	NQ - Rock Core, 1-7/8"	HSA - Hollow Stem Auger	RW - Rotary Wash
ST - Shelby Tube	CU - Cuttings	CFA - Continuous Flight Augers	RC - Rock Core
AWG - Rock Core, 1-1/8"	CT - Continuous Tube	DC - Driving Casing	

SCDOT Soil Test Boring Log

File No.:		Project No. (PIN):	1461-14-046	County:	Richland	Eng./Geo.:	S. Jones
Site Description:	I-77 Widening Design/Build Preparation					Route:	I-77
Boring No.:	B-01	Boring Location:	1215+74.4	Offset:	9.2' L	Alignment:	Mainline
Elev.:	337.9 ft	Latitude:	34.097467	Longitude:	80.956851	Date Started:	9/5/2014
Total Depth:	120 ft	Soil Depth:	120 ft	Core Depth:	N/A ft	Date Completed:	9/8/2014
Bore Hole Diameter (in):	4	Sampler Configuration		Liner Required:	Y (N)	Liner Used:	Y (N)
Drill Machine:	CME-45	Drill Method:	Rotary Wash	Hammer Type:	Automatic	Energy Ratio:	87%
Core Size:	N/A	Driller:	T. Miller	Groundwater:	TOB RW, N/A	24HR	15 ft

Elevation (ft)	Depth (ft)	MATERIAL DESCRIPTION	Graphic Log	Sample Depth (ft)	Sample No./Type	1st 6"	2nd 6"	3rd 6"	N Value	● SPT N VALUE ● PL MC LL ▲ FINES CONTENT (%)
82.0		CLAYEY SAND (SC) - COASTAL PLAIN - MIDDENDORF FORMATION , dense to very dense, wet, white, mostly fine to medium sand, some low to medium plasticity fines.								
252.9				83.5	SS-20	10	14	25	39	●
247.9		@ 88.5 feet, wet to saturated, white, gray, some low plasticity fines, trace fine gravel.		88.5	SS-21	12	16	19	35	●
242.9		@ 93.5 feet, wet, white, gray, light brown, some medium to high plasticity fines, absent fine gravel.		93.5	SS-22	9	16	21	37	●
237.9		@ 98.5 feet, very dense, wet, white, gray, brown, some medium plasticity fines.		98.5	SS-23	15	32	50/1"	100	●
232.9		@ 103.5 feet, wet to saturated, white, gray, brown, yellow, some low plasticity fines.		103.5	SS-24	15	30	32	62	●

LEGEND

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SAMPLER TYPE		DRILLING METHOD	
SS - Split Spoon	NQ - Rock Core, 1-7/8"	HSA - Hollow Stem Auger	RW - Rotary Wash
ST - Shelby Tube	CU - Cuttings	CFA - Continuous Flight Augers	RC - Rock Core
AWG - Rock Core, 1-1/8"	CT - Continuous Tube	DC - Driving Casing	

SC_DOT_14-046 I-77 BORING LOGS.GPJ SC_DOT.GDT 5/19/15

SCDOT Soil Test Boring Log

File No.:		Project No. (PIN):	1461-14-046	County:	Richland	Eng./Geo.:	S. Jones
Site Description:	I-77 Widening Design/Build Preparation					Route:	I-77
Boring No.:	B-01	Boring Location:	1215+74.4	Offset:	9.2' L	Alignment:	Mainline
Elev.:	337.9 ft	Latitude:	34.097467	Longitude:	80.956851	Date Started:	9/5/2014
Total Depth:	120 ft	Soil Depth:	120 ft	Core Depth:	N/A ft	Date Completed:	9/8/2014
Bore Hole Diameter (in):	4	Sampler Configuration		Liner Required:	Y (N)	Liner Used:	Y (N)
Drill Machine:	CME-45	Drill Method:	Rotary Wash	Hammer Type:	Automatic	Energy Ratio:	87%
Core Size:	N/A	Driller:	T. Miller	Groundwater:	TOB RW, N/A	24HR	15 ft

Elevation (ft)	Depth (ft)	MATERIAL DESCRIPTION	Graphic Log	Sample Depth (ft)	Sample No./Type	1st 6"	2nd 6"	3rd 6"	N Value	● SPT N VALUE ● PL MC LL ▲ FINES CONTENT (%)
227.9				108.5	SS-25	18	22	28	50	●
222.9		@ 113.5 feet, white, gray, yellow, brown, mostly fine to medium sand with trace coarse sand.		113.5	SS-26	27	43	50/5"	100	●
217.9	120.0	@ 118.5 feet, medium dense, saturated, some medium plasticity fines, medium plasticity nodules.		118.5	SS-27	10	8	13	21	●
		Boring Terminated at 120 feet.								
212.9										
207.9										

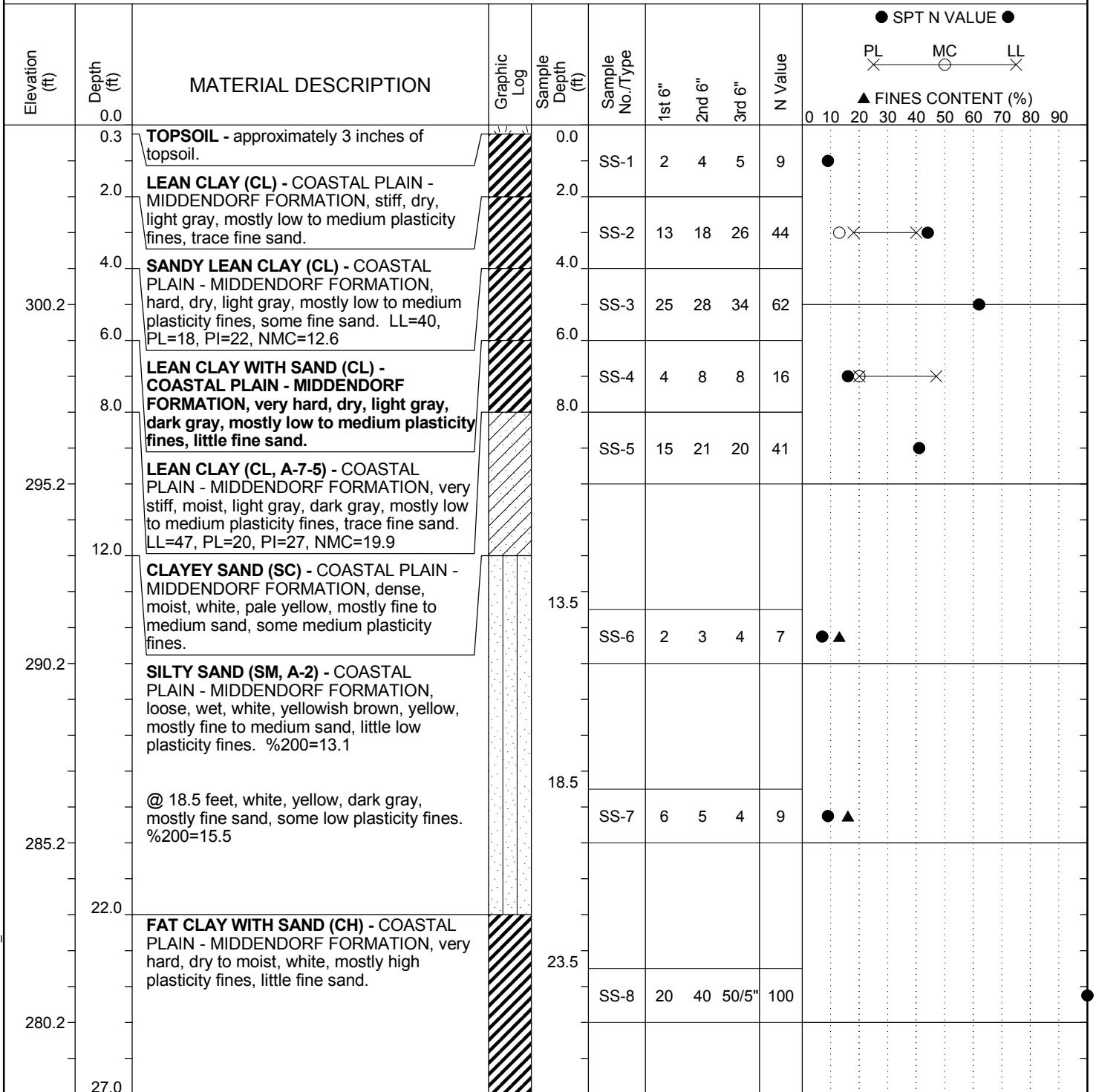
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SAMPLER TYPE		DRILLING METHOD	
SS - Split Spoon	NQ - Rock Core, 1-7/8"	HSA - Hollow Stem Auger	RW - Rotary Wash
ST - Shelby Tube	CU - Cuttings	CFA - Continuous Flight Augers	RC - Rock Core
AWG - Rock Core, 1-1/8"	CT - Continuous Tube	DC - Driving Casing	

SC_DOT_14-046 I-77 BORING LOGS.GPJ SC_DOT.GDT 5/19/15

SCDOT Soil Test Boring Log

File No.:		Project No. (PIN):	1461-14-046	County:	Richland	Eng./Geo.:	S. Jones
Site Description:	I-77 Widening Design/Build Preparation					Route:	I-77
Boring No.:	B-02	Boring Location:	1217+31.7	Offset:	13.4' R	Alignment:	Mainline
Elev.:	305.2 ft	Latitude:	34.067554	Longitude:	80.922215	Date Started:	9/10/2014
Total Depth:	100 ft	Soil Depth:	100 ft	Core Depth:	N/A ft	Date Completed:	9/10/2014
Bore Hole Diameter (in):	4	Sampler Configuration		Liner Required:	Y (N)	Liner Used:	Y (N)
Drill Machine:	CME-55	Drill Method:	Rotary Wash	Hammer Type:	Automatic	Energy Ratio:	81%
Core Size:	N/A	Driller:	T. Miller	Groundwater:	TOB	RW, N/A	24HR Filled ATD



LEGEND

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SAMPLER TYPE		DRILLING METHOD	
SS - Split Spoon	NQ - Rock Core, 1-7/8"	HSA - Hollow Stem Auger	RW - Rotary Wash
ST - Shelby Tube	CU - Cuttings	CFA - Continuous Flight Augers	RC - Rock Core
AWG - Rock Core, 1-1/8"	CT - Continuous Tube	DC - Driving Casing	

SCDOT Soil Test Boring Log

File No.:		Project No. (PIN):	1461-14-046	County:	Richland	Eng./Geo.:	S. Jones
Site Description:	I-77 Widening Design/Build Preparation					Route:	I-77
Boring No.:	B-02	Boring Location:	1217+31.7	Offset:	13.4' R	Alignment:	Mainline
Elev.:	305.2 ft	Latitude:	34.067554	Longitude:	80.922215	Date Started:	9/10/2014
Total Depth:	100 ft	Soil Depth:	100 ft	Core Depth:	N/A ft	Date Completed:	9/10/2014
Bore Hole Diameter (in):	4	Sampler Configuration		Liner Required:	Y (N)	Liner Used:	Y (N)
Drill Machine:	CME-55	Drill Method:	Rotary Wash	Hammer Type:	Automatic	Energy Ratio:	81%
Core Size:	N/A	Driller:	T. Miller	Groundwater:	TOB	RW, N/A	24HR
							Filled ATD

Elevation (ft)	Depth (ft)	MATERIAL DESCRIPTION	Graphic Log	Sample Depth (ft)	Sample No./Type	1st 6"	2nd 6"	3rd 6"	N Value	● SPT N VALUE ● PL MC LL ▲ FINES CONTENT (%)
275.2		CLAYEY SAND (SC, A-2-6) - COASTAL PLAIN - MIDDENDORF FORMATION, very dense, moist to wet, white, yellow, mostly fine to medium sand, some medium to high plasticity fines.		28.5	SS-9	21	29	35	64	●
270.2		@ 33.5 feet, medium dense, white, pale yellow, some medium plasticity fines.		33.5	SS-10	8	9	10	19	●
265.2		@ 38.5 feet, dense, wet, white, gray, yellow, some low plasticity fines, nodules of medium plasticity fines. LL=27, PL=13, PI=14, NMC=19.5, %200=17.2		38.5	SS-11	12	19	21	40	●
260.2		SANDY FAT CLAY (CH) - COASTAL PLAIN - MIDDENDORF FORMATION, very hard, moist to wet, gray, yellowish brown, mostly high plasticity fines, some fine to medium sand with trace coarse sand, trace fine gravel.		43.5	SS-12	16	35	40	75	●
255.2		CLAYEY SAND (SC) - COASTAL PLAIN - MIDDENDORF FORMATION, dense, moist, grayish white, yellow, mostly fine to medium sand, some medium to high plasticity fines.		48.5	SS-13	14	20	26	46	●
				53.5						

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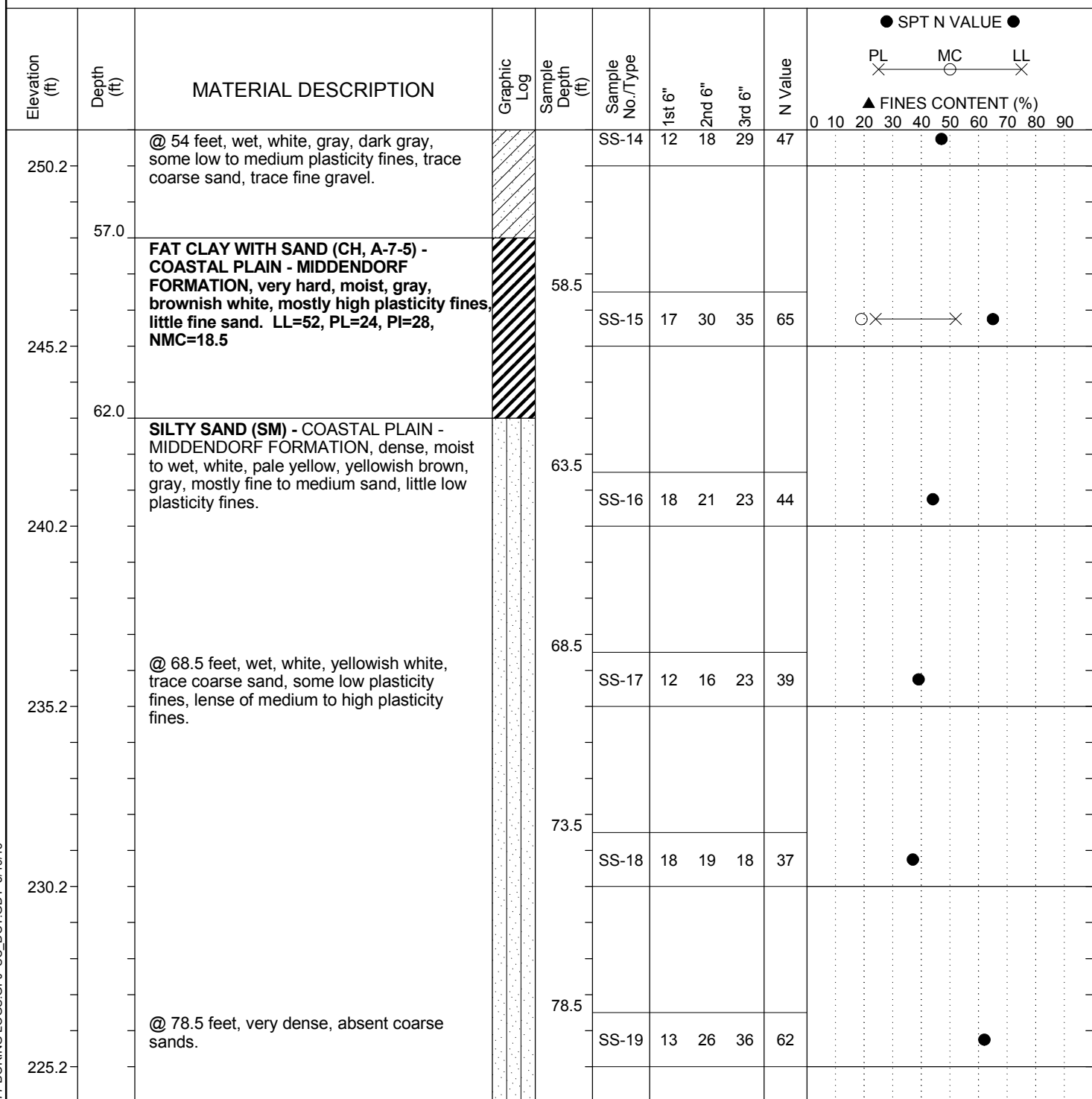
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SAMPLER TYPE		DRILLING METHOD	
SS - Split Spoon	NQ - Rock Core, 1-7/8"	HSA - Hollow Stem Auger	RW - Rotary Wash
ST - Shelby Tube	CU - Cuttings	CFA - Continuous Flight Augers	RC - Rock Core
AWG - Rock Core, 1-1/8"	CT - Continuous Tube	DC - Driving Casing	

SC_DOT_14-046 I-77 BORING LOGS.GPJ SC_DOT.GDT 5/19/15

SCDOT Soil Test Boring Log

File No.:		Project No. (PIN):	1461-14-046	County:	Richland	Eng./Geo.:	S. Jones
Site Description:	I-77 Widening Design/Build Preparation					Route:	I-77
Boring No.:	B-02	Boring Location:	1217+31.7	Offset:	13.4' R	Alignment:	Mainline
Elev.:	305.2 ft	Latitude:	34.067554	Longitude:	80.922215	Date Started:	9/10/2014
Total Depth:	100 ft	Soil Depth:	100 ft	Core Depth:	N/A ft	Date Completed:	9/10/2014
Bore Hole Diameter (in):	4	Sampler Configuration		Liner Required:	Y (N)	Liner Used:	Y (N)
Drill Machine:	CME-55	Drill Method:	Rotary Wash	Hammer Type:	Automatic	Energy Ratio:	81%
Core Size:	N/A	Driller:	T. Miller	Groundwater:	TOB	RW, N/A	24HR Filled ATD



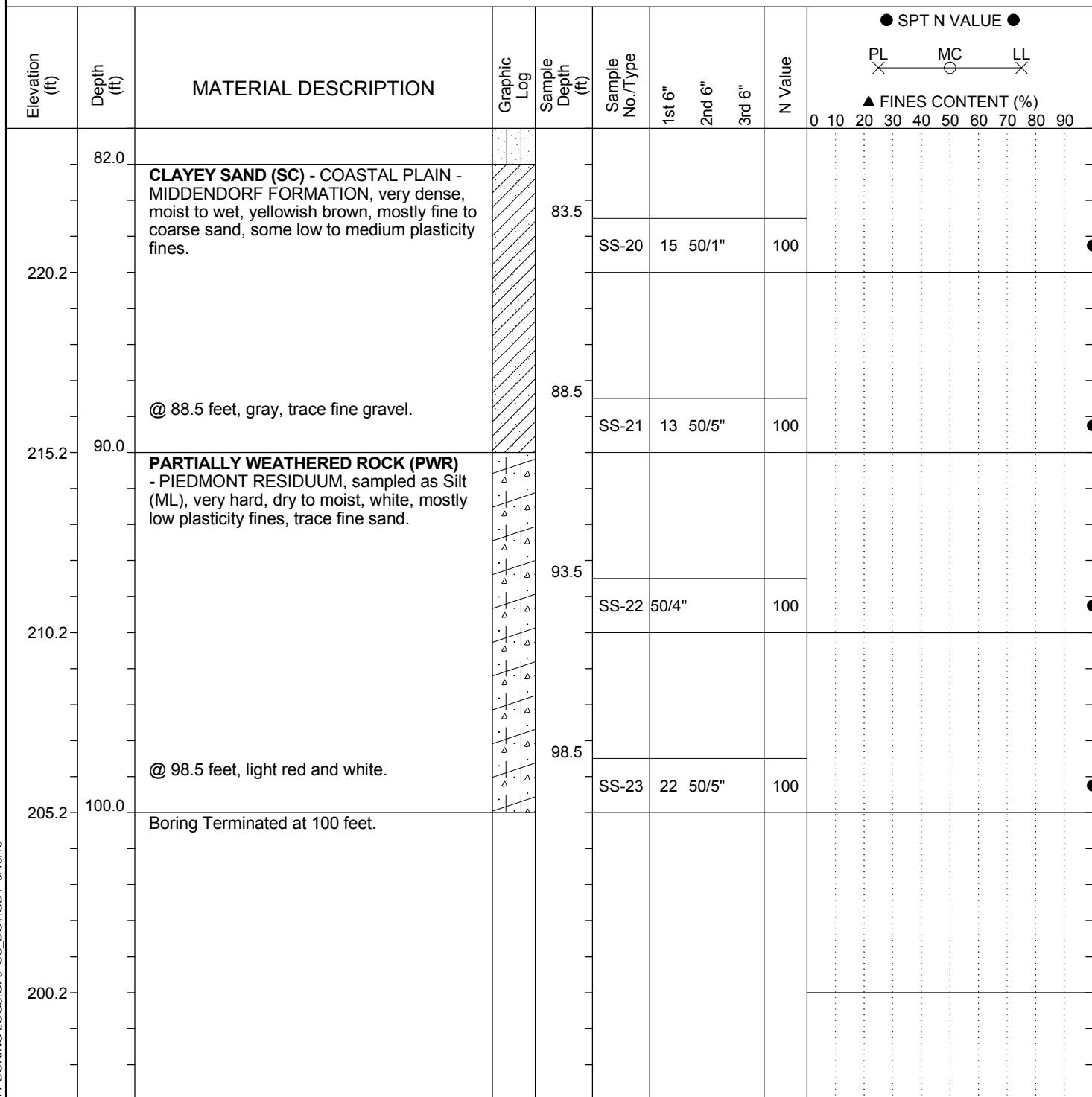
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SAMPLER TYPE		DRILLING METHOD	
SS - Split Spoon	NQ - Rock Core, 1-7/8"	HSA - Hollow Stem Auger	RW - Rotary Wash
ST - Shelby Tube	CU - Cuttings	CFA - Continuous Flight Augers	RC - Rock Core
AWG - Rock Core, 1-1/8"	CT - Continuous Tube	DC - Driving Casing	

SCDOT Soil Test Boring Log

File No.:		Project No. (PIN):	1461-14-046	County:	Richland	Eng./Geo.:	S. Jones
Site Description:	I-77 Widening Design/Build Preparation					Route:	I-77
Boring No.:	B-02	Boring Location:	1217+31.7	Offset:	13.4' R	Alignment:	Mainline
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Total Depth:	100 ft	Soil Depth:	100 ft	Core Depth:	N/A ft	Date Completed:	9/10/2014
Bore Hole Diameter (in):	4	Sampler Configuration		Liner Required:	Y (N)	Liner Used:	Y (N)
Drill Machine:	CME-55	Drill Method:	Rotary Wash	Hammer Type:	Automatic	Energy Ratio:	81%
Core Size:	N/A	Driller:	T. Miller	Groundwater:	TOB	RW, N/A	24HR Filled ATD



LEGEND

SAMPLER TYPE		DRILLING METHOD	
SS - Split Spoon	NQ - Rock Core, 1-7/8"	HSA - Hollow Stem Auger	RW - Rotary Wash
ST - Shelby Tube	CU - Cuttings	CFA - Continuous Flight Augers	RC - Rock Core
AWG - Rock Core, 1-1/8"	CT - Continuous Tube	DC - Driving Casing	

SC_DOT_14-046 I-77 BORING LOGS.GPJ SC_DOT.GDT 5/19/15